

Brain Atlas Exploration and Visualization tool

GROUP 216

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PROBLEM STATEMENT

- Neuroscience researchers studying the mouse brain face three practical problems:
 1. The Allen CCFv3 reference atlas is distributed as raw 3D volume files and CSV metadata that are difficult to explore without specialized software.
 2. Mapping a clicked voxel to its full anatomical hierarchy requires joining multiple metadata tables by hand
 3. When a researcher identifies an unfamiliar region, they must leave the atlas and search PubMed separately to find relevant literature. Our project solves all three by combining visual exploration, hierarchical lookup, and AI-assisted literature access in one interface.

OVERVIEW

- Build a web-based interactive explorer for the Allen Mouse Brain Common Coordinate Framework v3 (CCFv3), enabling neuroscience researchers to visually browse a 10 μm resolution reference atlas, identify any brain region by clicking on it, highlight regions at any level of the anatomical hierarchy, and ask natural-language questions about regions to an AI assistant grounded in both a curated knowledge base and live research paper summaries.
- Scope: 1,320 $\times$ 800 $\times$ 1,140 voxel annotation volume, three orthogonal views (sagittal/coronal/transverse), full Allen ontology tree, dual-AI chatbot (local Ollama + OpenAI), and integration with the CortexMap research API.

TECHNOLOGIES USED

- Backend (Python): Flask, Flask-CORS, NumPy, SciPy, SimpleITK, Pillow, Matplotlib, abc_atlas_access AI / RAG: LangChain, ChromaDB, HuggingFace Embeddings (all-MiniLM-L6-v2), Ollama (gemma3:4b), OpenAI GPT-4o-mini
- Frontend: HTML5, CSS3, Vanilla JavaScript, Canvas rendering, Google Fonts (JetBrains Mono, Syne) Data Sources: Allen CCF-2020 dataset (~400 MB), CortexMap API (capstone.ssdd.dev) Dev Tools: Git, Python 3, Node.js for asset tooling

TARGET USERS

- Neuroscience researchers at Penn State College of Medicine and affiliated labs who need to quickly locate and understand mouse brain regions
- Graduate students learning mouse neuroanatomy who benefit from interactive exploration over static textbook figures
- Computational neuroscientists who need programmatic access to parcellation indices, colors, and hierarchy for their own pipelines
- The CortexMap research team, whose paper-summary orchestrator is consumed directly by our chat backend

INTRODUCTION - MEET THE TEAM

Muhammad Danish Zahin Bin Rafizal

Atlas Visualization and Backend

- Implemented Allen CCFv3 atlas data loading
- Built slice rendering (Sagittal, Coronal, Transverse)
- Developed voxel lookup and region identification
- Implemented region highlighting and ontology grouping
- Created backend API endpoints for atlas interaction

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Frontend UI/UX and Interaction

- Designed the three-view atlas interface
- Built interactive slice sliders and view controls
- Implemented region search and inspector panel
- Developed hierarchy browser and coordinate display
- Integrated AI assistant panel into the interface

Alif Danial Kamarole Erifin

AI and Brain Knowledge System

- Implemented the Brain Atlas Assistant
- Integrated AI models (Ollama + GPT) for region questions
- Built region knowledge base and CortexMap enrichment
- Developed chat interaction with atlas regions
- Connected atlas selections to AI explanations

SYSTEM ARCHITECTURE OVERVIEW

The system has three main layers communicating through a REST API:

- Data Layer — Allen CCF-2020 image volumes and metadata CSVs, plus our curated brain_regions_kb.json
- Backend Layer (App.py) — Flask server that loads and slices the 3D atlas, handles lookups and highlighting, implements smart search, and routes chat queries through the RAG pipeline
- Frontend Layer (index.html) — single-page web app that renders the atlas canvases, ontology tree, search, and chat panel

External integrations: CortexMap API for live research summaries, OpenAI API for GPT-based answers, and the Allen Brain Atlas API for the ontology tree.

DATA LOADING PIPELINE

When the server starts, it loads all the brain atlas data it needs to run:

- Downloads the atlas files (about 400 MB total) from the Allen Brain Atlas. If the files are already saved on disk, it just uses the cached copies instead of re-downloading.
- Loads three 3D brain images into memory:
 - The annotation volume — tells us which brain region each voxel belongs to
 - The template volume — the grayscale anatomical picture of the brain
 - The boundary volume — the outlines between regions
- Loads five reference tables that describe every region: its name, its acronym, its color, which groups it belongs to, and the full parent-child hierarchy tree.
- Builds fast lookup tables in memory so that when a user clicks a region, the server can instantly find its name, color, and place in the hierarchy without searching through the tables every time.

This all happens once at startup, so the app stays fast while users are using it.

ATLAS SLICING & RENDERING

The 3D brain is sliced into 2D images in three orientations:

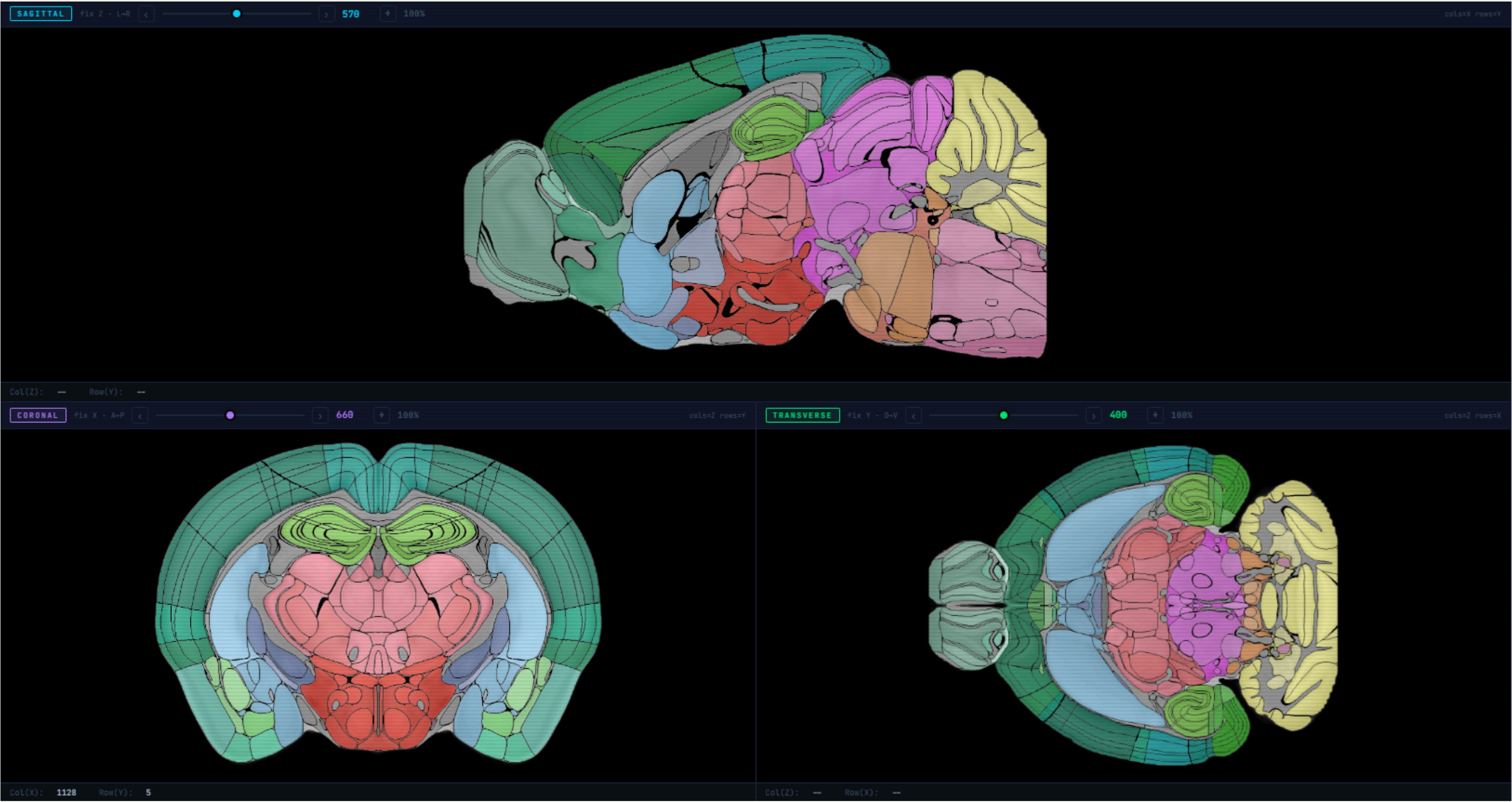
- Sagittal — side view (left-to-right slicing)
- Coronal — front view (front-to-back slicing)
- Transverse — top view (top-to-bottom slicing)

For each slice, the server layers three things together:

1. The grayscale anatomy as the base
2. The colored regions on top (using the pre-built LUT)
3. The boundary lines between regions for clarity

The final image is sent to the browser as a PNG.

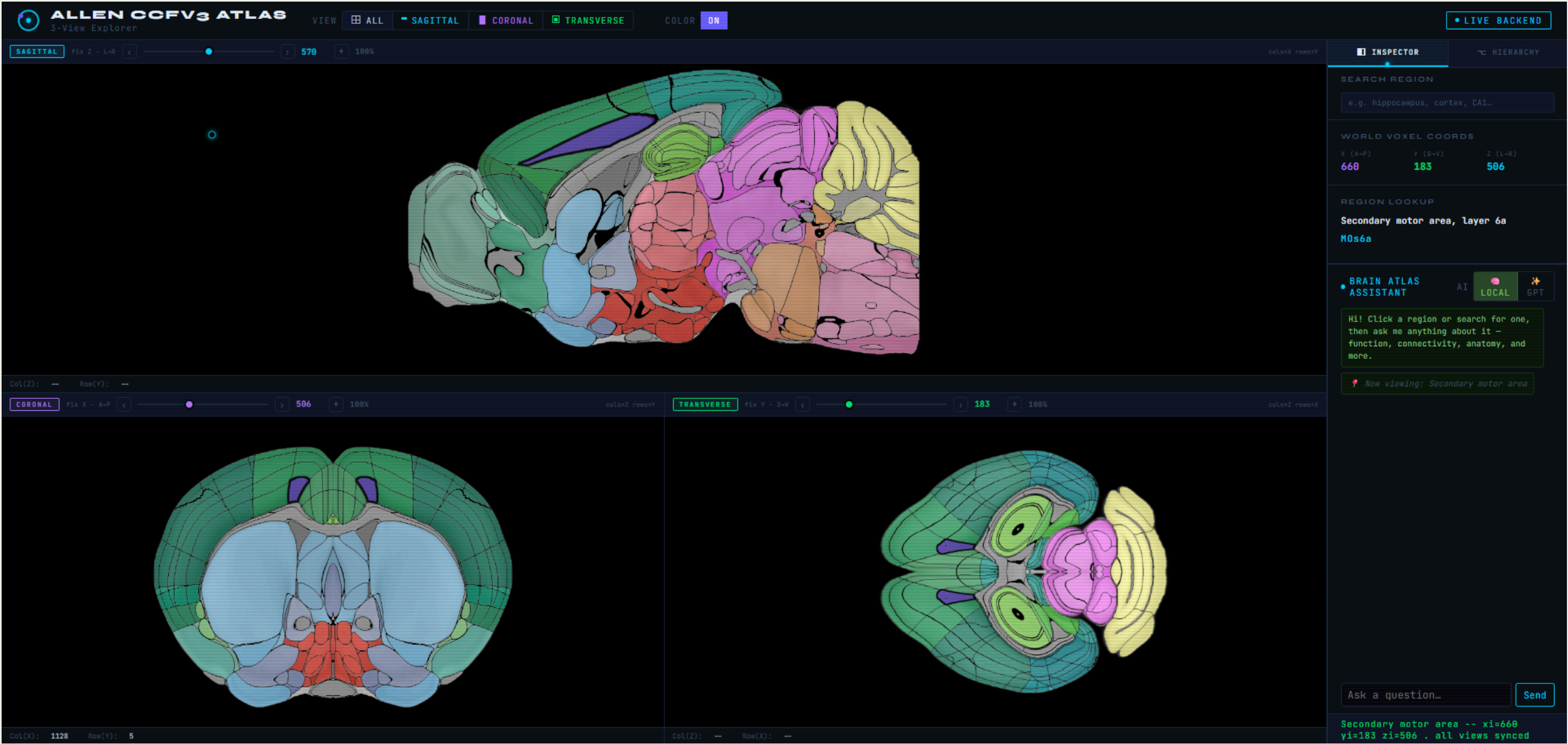
PICTURE



CLICK-TO-IDENTIFY PIPELINE

- When a user clicks on the atlas, the frontend sends the click coordinates to the /api/lookup endpoint. The server reads the parcellation index at that voxel, then joins it with the name, color, and acronym tables.
- Then pass this to AI chatbot so they can see the clicked region

PICTURE



REGION HIGHLIGHTING

The `/api/highlight` endpoint generates a colored overlay mask for any selected region:

- Interior fill — the region's atlas color at ~70% opacity
- Thick outline — drawn in near-black using scipy dilation + erosion so the boundary is clearly visible

A second endpoint, `/api/highlight_indices`, highlights multiple regions at once — used for big container groups like "Cerebrum" that cover hundreds of sub-regions.

ONTOLOGY TREE

The atlas has a parent-child hierarchy of brain regions, built from the Allen parcellation term tree.

- The `_collect_leaf_indices()` function walks down from any parent node and collects all its descendant regions
- So clicking "Cerebrum" highlights every one of its ~1,200 leaf regions in one action

The full tree is fetched from ABatlas

PICTURE



SIDEBAR

Sidebar consists of two sections:

- Search Bar
- Inspector: - Chatbot with options of models
- Data regarding highlighted brain region
- Hierarchy: Anatomical hierarchy



DUAL-AI CHATBOT

Users can switch between two AI backends from the frontend:

- Local Ollama (gemma3:4b) — Full RAG pipeline: our knowledge base is split into chunks, embedded with all-MiniLM-L6-v2, and stored in a ChromaDB vector store for semantic search
- OpenAI (GPT-4o-mini) — No local KB needed; the server builds region context from the atlas tables and sends it to OpenAI's API with a neuroscientist system prompt

C O R T E X M A P - F I R S T R A G

When a question is asked, the chatbot follows this decision tree:

1. Call CortexMap — look up the region's UUID and fetch live research paper summaries
2. If summaries exist → the LLM answers using only those summaries, and the UI shows clickable PMC links
3. If CortexMap fails or has no data → fall back to our local brain_regions_kb.json
4. Return a status code (enriched, generating, unreachable, etc.) + a user-friendly notice

PICTURE

ALLEN CCFV3 ATLAS
3-View Explorer

VIEW

ALL

SAGITTAL

CORONAL

TRANSVERSE

COLOR

ON

SAGITTAL

fix Z - L-R

283

100%

CORONAL

fix X - A-P

587

100%

TRANSVERSE

fix Y - D-V

327

100%

INSPECTOR

HIERARCHY

SEARCH REGION

e.g. hippocampus, cortex, CA1...

WORLD VOXEL COORDS

X (A-P)283

Y (D-V)327

Z (L-R)587

REGION LOOKUP

Caudoputamen

CP

BRAIN ATLAS ASSISTANT

AI

LOCAL

GPT

Switched to Local Dllama RAG. Ask away!

Now viewing: Caudoputamen

tell me about this area

Local

The caudoputamen, often referred to as the striatum, is a critical brain region located in the basal ganglia, situated deep within the cerebral hemispheres. It comprises the caudate nucleus and the putamen and plays a vital role in motor control, cognitive processes, reward-related behaviors, and the modulation of voluntary movement and the reinforcement of learning. It's composed of medium spiny neurons and various interneurons, exhibiting a complex organization with distinct regions corresponding to different functional roles. It receives inputs from the cortex, thalamus, and midbrain dopaminergic neurons, and sends outputs primarily to the globus pallidus and substantia nigra. The caudoputamen is involved in motor control, learning, and reward processing, interacts with the prefrontal cortex and limbic structures, and is significantly affected in disorders like Parkinson's Disease, Huntington's Disease, and Alzheimer's Disease, leading to symptoms such as cognitive deficits, motor impairments, and behavioral changes. Current research explores innovative treatments targeting this region, including focused ultrasound and neuroprotective agents.

PubMed Sources (ContextMap):

PMC13063251 /PMC13070663 /PMC13036399 /PMC13066316 /PMC13069099 /PMC13069889 /PMC13065517 /PMC13070112 /PMC13070256 /PMC13070641 /PMC13070226 /PMC13070227 /PMC13070126 /PMC13070629 /PMC13068870 /PMC13070614 /PMC13070304 /PMC13070411 /PMC13069501 /PMC13069362 /PMC13069153 /PMC13069703 /PMC13069066 /PMC13069807 /PMC13068512 /PMC13069952 /PMC13069653 /PMC13068066 /PMC13068134 /PMC13066547 /PMC13067576 /PMC13067832 /PMC13068360 /PMC13067311 /PMC13066286 /PMC13063209 /PMC13069523 /PMC13069442 /PMC13068684 /PMC13068213 /PMC13055200 /PMC13054001 /PMC13055387 /PMC13053744 /PMC13054538 /PMC13053294 /PMC13055079 /PMC13055836 /PMC13055314 /PMC13054074 /PMC13055203 /PMC13054540 /PMC13054064 /PMC13055450

Ask a question...

Go to Settings to activate Windows.

Send

Caudoputamen -- xi=283 yi=327 zi=587 . all views synced

GPT RESPONSE PARSER

When a user asks something like "List all areas connected to pain," GPT returns a natural-language answer containing region names. A parser then extracts those region names from GPT's text so the atlas can highlight them.

How it works:

1. User asks a list-style question through the OpenAI chatbot
2. GPT returns an answer mentioning multiple brain regions by name or acronym (e.g., "The thalamus, anterior cingulate cortex, and periaqueductal gray are all involved...")
3. The parser scans GPT's response and matches each mentioned region against our atlas name and acronym tables
4. Matched regions are converted into their parcellation indices
5. The frontend highlights all of them at once on the atlas using `/api/highlight_indices` which makes them clickable if they exist and not clickable if not exist

PICTURE

BRAIN ATLAS ASSISTANT

AI

LOCAL

GPT

Switched to ChatGPT (GPT-4o-mini). Ask away!

list me the parts that relate to pain

GPT

The Basolateral Amygdalar Nucleus, Anterior Part (BLAa) is involved in the processing of emotional responses, including those related to pain. Here are some key brain regions connected to pain processing that may interact with the BLAa:

1. ****Anterior Cingulate Cortex (ACC)**** - Involved in the emotional aspect of pain.

2. ****Insular Cortex (IC)**** - Processes the sensory-discriminative aspects of pain.

3. ****Periaqueductal Gray (PAG)**** - Plays a crucial role in pain modulation and descending pain control pathways.

4. ****Nucleus Accumbens (NAc)**** - Involved in the motivational aspects of pain and reward.

5. ****Hippocampus (HP)**** - Although primarily associated with memory, it can influence pain perception through contextual processing.

These regions are interconnected with the BLAa and contribute to the complex neural circuitry that processes pain in the mouse brain.

SEMANTIC REGION MAP

3 / 6 regions

Ask a question...

Send

"Nucleus accumbens" -- xi=476 yi=592 zi=423 . 4,401,366 voxels

SEMANTIC REGION MAP

3 / 6 regions

list me the parts that relate to pain

Anterior Part *not in volume*

Anterior cingulate area, dorsal part (5)

Insular Cortex *not in volume*

Periaqueductal gray (5)

Nucleus accumbens

Hippocampus *not in volume*

↑ click any chip to navigate atlas

Now viewing: Nucleus accumbens

FUNCTIONAL SUMMARY

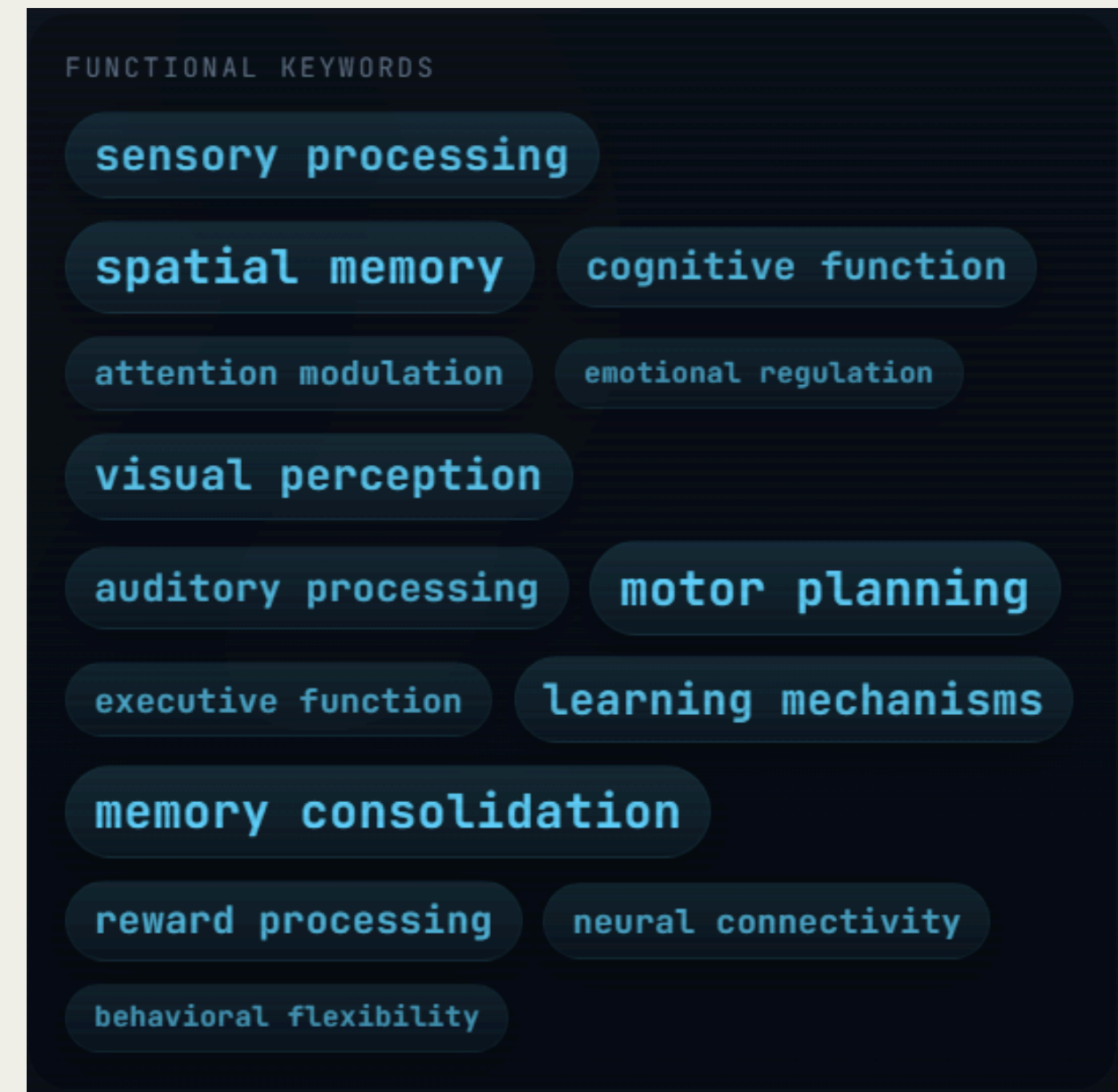
- The function extracts data related to the brain region's function, connectivity, clinical relevance, and notes to build a concise summary.
- It prioritizes the most important aspects, combining information from key fields like function and connectivity to provide a brief but informative description.
- The data is cleaned and split into sentences, removing redundant information and ensuring clarity for presentation.

FUNCTIONAL SUMMARY

The lateral dorsal nucleus of the thalamus (LD) is involved in processing and relaying sensory information, particularly related to spatial learning and memory. It connects with the hippocampus and other limbic structures, playing a crucial role in integrating sensory inputs with emotional and cognitive functions. This connectivity supports its involvement in navigation and the formation of spatial representations.

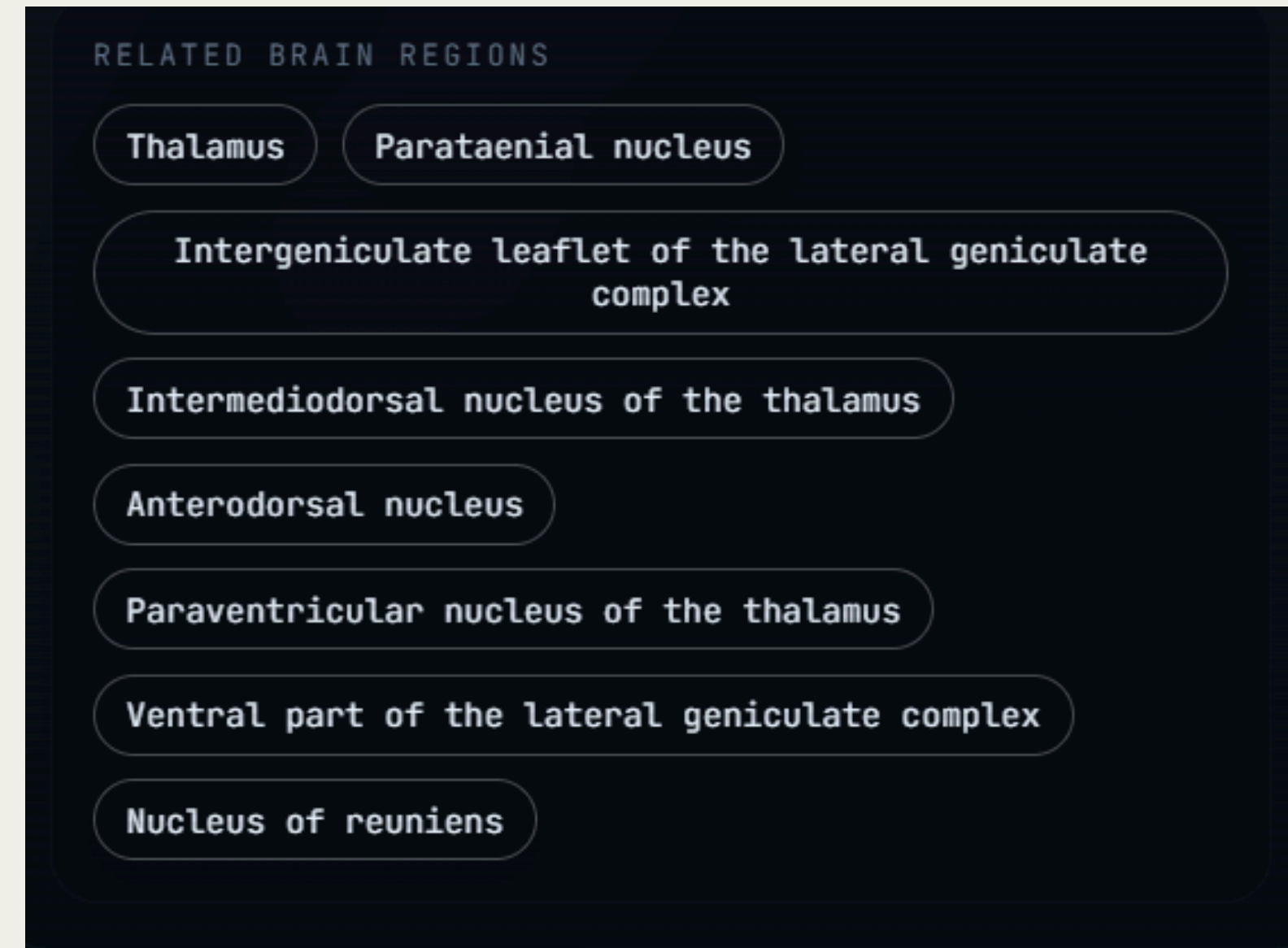
FUNCTIONAL KEYWORDS

- The function analyzes data related to the brain region's function, connectivity, clinical relevance, and notes.
- It extracts keywords associated with the region, focusing on the most relevant ones.
- These keywords are weighted based on their frequency or relevance to the region's characteristics.
- The function then returns the top keywords, helping users quickly understand the region's function and relevance.



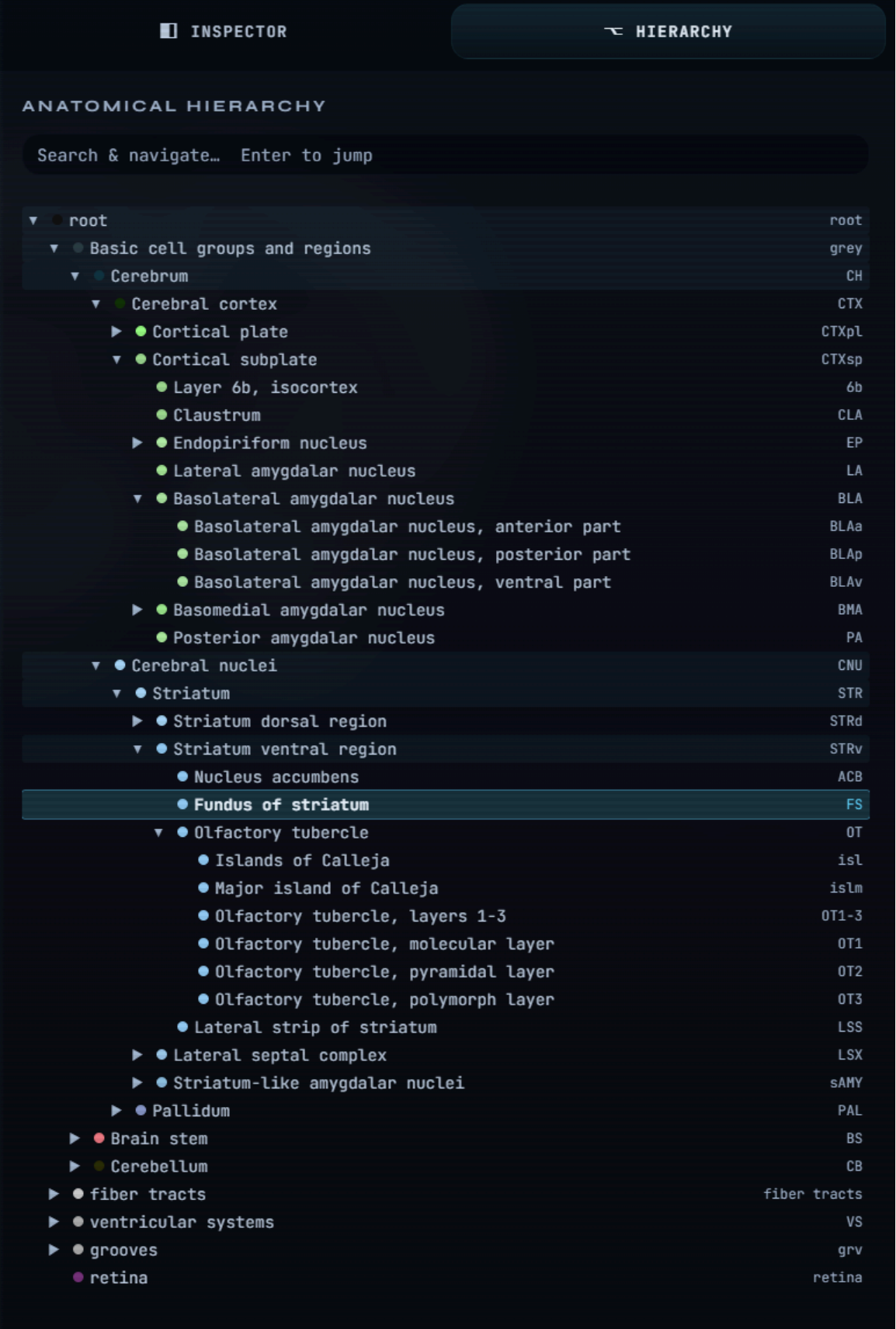
RELATED REGIONS

- The function examines the brain region's connectivity data, looking for relationships like inputs and outputs with other regions.
- It identifies related regions by matching these connections with other brain areas sharing similar functions or anatomical links.
- The function then returns a list of related regions, enabling users to explore connected or functionally similar areas of the brain.



ANATOMICAL HIERARCHY

The Anatomical Hierarchy organizes brain regions in a clear, easy-to-navigate structure. It allows users to quickly find and explore related regions with minimal effort, displaying relevant details for each area, making the system user-friendly and efficient.



IMPLEMENTATION CHALLENGES

- Loading and handling very large atlas volumes efficiently, since the Allen CCFv3 data includes full 3D annotation, template, and boundary arrays.
- Correctly mapping the 3D atlas axes into Sagittal, Coronal, and Transverse 2D slice views without orientation mistakes.
- Mapping user clicks to the right voxel, identifying the matching region, and rendering highlights or grouped regions correctly.
- Integrating the AI assistant with atlas data, connecting region selection, the JSON knowledge base, local/OpenAI models, and CortexMap enrichment into one usable chat workflow

Demonstrations!

FUTURE PLANS

Human Brain Atlas Integration - Extend the platform to support human brain datasets

Advanced 3D Visualization - Expand from slice-based views to full interactive 3D models, enabling users to rotate, zoom, and inspect anatomical regions in three dimensions.

Enhanced AI Knowledge Integration - improve the AI assistant to provide deeper explanations of anatomical functions, connectivity, and research insights directly from atlas regions.

CONCLUSION

- We built an interactive brain atlas exploration tool using the Allen CCFv3 dataset.
- We integrated multi-view visualization (sagittal, coronal, transverse) with region lookup and highlighting.
- We combined atlas data, backend APIs, and an AI assistant to enable interactive exploration of brain regions.
- We demonstrates how complex neuroscience datasets can be made accessible through modern visualization tools.

This project shows how complex neuroscience data can be made more accessible and interactive which would support learning and research exploration.

Thank you!

OPEN FOR Q&A